DATABASE MANAGEMENT SYSTEM (DBMS) PROJECT PROPOSAL

Pharmacy Management System

Technology: MySQL

Submitted by:		**Student ID:**	
Course / Semester:		**Date:**	

1. Project Overview

The Pharmacy Management System is a database-driven application that manages the daily work of a pharmacy: medicines, suppliers, customers, purchases, sales and stock.

Problem it solves: many small pharmacies still keep records on paper or in scattered files, which causes stock mistakes, expired or out-of-stock medicines, and slow searching. This project stores everything in one organized MySQL database so records are accurate, easy to find and quick to report.

2. Main Features

Feature	Description
Medicine Management	Add, update, delete and search medicines (name, category, price, expiry date)
Supplier Management	Keep supplier details and contact information
Customer Management	Store customer records and their purchase history
Purchase Management	Record medicines bought from suppliers
Sales Management	Record medicines sold to customers and generate bills
Stock Management	Automatically update stock after every purchase and sale
Basic Reports	Sales, purchase, low-stock and expiry reports

3. Database

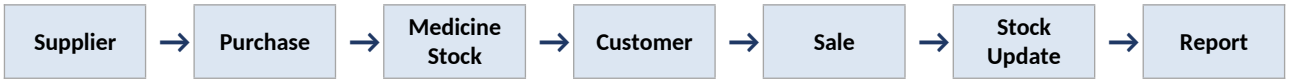
The project will use MySQL as the database. Proposed main tables:

Table	Primary Key	Foreign Key(s)	Purpose
categories	category_id	—	Medicine categories (e.g. tablet, syrup)
medicines	medicine_id	category_id → categories	Medicine details, price and stock quantity
suppliers	supplier_id	—	Supplier information
customers	customer_id	—	Customer information
purchases	purchase_id	supplier_id → suppliers	Purchase orders (date, total)
purchase_items	purchase_item_id	purchase_id, medicine_id	Medicines and quantities in each purchase
sales	sale_id	customer_id → customers	Sales / bills (date, total)
sale_items	sale_item_id	sale_id, medicine_id	Medicines and quantities in each sale

Relationships:

- One category has many medicines; one supplier has many purchases; one customer has many sales (one-to-many).
- One purchase has many purchase_items and one sale has many sale_items (one-to-many).
- Medicines ↔ Purchases and Medicines ↔ Sales are many-to-many, connected through purchase_items and sale_items.

4. Basic Workflow



5. DBMS Concepts

ER Diagram	**Relational Database**	**Primary Key**	**Foreign Key**
Normalization	**CRUD**	**SQL JOIN**	**Basic Queries**

6. Technology Stack

Frontend	Backend	Database	Local Server
HTML, CSS, JavaScript	PHP	MySQL	XAMPP

7. Project Goal

- Design a well-structured, normalized MySQL database for a pharmacy.
- Replace manual record keeping with a fast and reliable digital system.
- Keep medicine stock accurate through automatic updates after purchases and sales.
- Apply core DBMS concepts (keys, relationships, JOINs, CRUD) in a practical project.

8. Approval

Teacher's Approval

☐ Approved
☐ Changes Required
☐ Not Approved

Teacher's Comments:

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Signature: _____ **Date:** _____